

2026 Robocup challenge – Objective, requirements, and rules

Note – these rules are subject to update or modification

Version 1.0 (23/02/2026) – changes since last version shown in red

Scenario: This year, the Robocup continues its Quidditchy twist

Objective: You have **2 minutes** to gain the highest score by collecting **target weights** and/or **snitches** while avoiding **dummy weights** and **bludgers** using your mobile robot.

Competition: Two robots will be in the arena at the same time, collecting *target weights* for their team. There will be a number of *weights* in the arena for each round, as well as *mobile elements* (*snitches* and *bludgers*).

Target weights and *snitches* add to your score. *Dummy weights* and *bludgers* subtract from your score.

Weights and *mobile elements* are considered collected, and thus contribute to your final score, if they are on board your robot at the end of the round. To make matters more interesting, **at the end of the round, your robot can only have a maximum of 3 weights on board.**

Target weights can be **delivered back to your home base**, and you **receive a bonus** for achieving this.

At the end of the 2 minutes, the team with the highest score will be declared the winner.

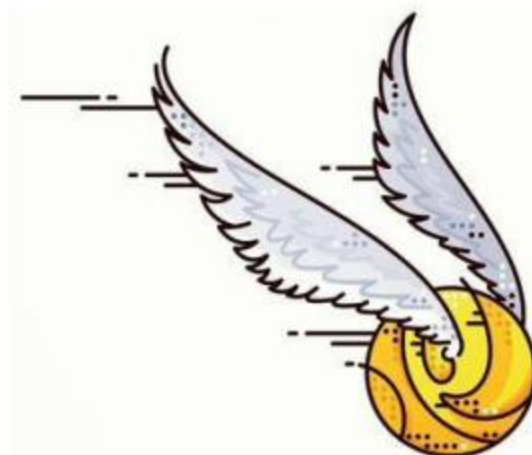
The score for each round is calculated as:

$\text{Score} = 1x (\text{weight of target weights on board robot}) + 2x (\text{weight of target weights in home base at the end of the round}) + 3x (\text{number of snitches on board robot}) - 3x (\text{number of bludgers on board robot}) - 0.5x (\text{number of dummy weights on board robot}) - \text{any penalties (see the fine print)}$

For example, if at the end of the round, your robot has:

- 0.75 kg target weight on board
- 3 kgs of target weights in your home base
- 1 snitch on board
- 1 dummy weight on board

$\text{Score} = 1x 0.75 + 2x 3 + 3x 1 - 0.5x 1 = 9.25 \text{ points.}$



Environment: The arena in which the race takes place is 2.4 x 4.9m with 400mm high walls. The arena will contain a **number of obstacles** that your robot will need to avoid or negotiate. The **position of the obstacles** for the final competition will not be revealed until the day of competition and **will change regularly during the competition rounds**, so your robot must be able to deal with these unknowns.

Within the arena, there will be a number of *target weights* (> 5) and *dummy weights*. The shape of all *weights* will be identical (i.e. A cylinder of diameter 50mm and height 70mm with an annular groove to facilitate gripping) but vary in colour and weight. The location of the weights will vary and you **will not know these positions ahead of time**.

There may also be up to 2 *mobile elements* (*snitches* and/or *bludgers*) consisting of plastic Sphero Bolt robots (<https://www.sphero.com/sphero-bolt>), approximately 73mm in diameter. They will roam the arena at up to 7 m/s and will be differentiated by the colour of their illuminated LEDs: *snitches* will be illuminated yellow (gold) and *bludgers* red.

Within the arena there are two areas designated as bases. These areas will be coloured *green* and *blue*, different from the surrounding arena (black) and obstacles (red). These bases will measure 600mm x 600mm and include a low rim to prevent weights rolling out. The robots will start each round on one of these bases.

Hardware: You must construct your robot using:

Controller: Teensy 4.0 (ARM Cortex-M7 microcontroller). You cannot use other boards/controllers – but you won't need to cos a 32-bit processor running at up to 600 MHz is more than enough!

Chassis: You have the choice of using a supplied tracked chassis or building your own custom chassis from scratch.



Sensors/actuators/structure: You will be provided with:

- i. A box of parts including various sensors, actuators, and structural elements such as aluminium plate, rods, acrylic etc.
- ii. A budget of \$50 which you can use to order additional components.
- iii. 500g of PLA for 3D printing. Additional material will come out of your \$50 budget at 5c/g.

Constraints: To add an element of freshness, this year rubber bands and other elastomeric elements (e.g. rubber strips or strings) are banned (band?) and cannot form part of your robot design (see clause 1.6 of the fine print for details).

Technical support: Julian Murphy

The fine print:

1. Robots

- 1.1. Robots **must be autonomous**. There can be no human intervention in the robot's operation, either physically or via software during the competition.
- 1.2. Robots must have a defined **front end** that will be used for aligning the start direction.
- 1.3. Robots must be controlled by the supplied **Teensy 4.0** (ARM Cortex-M7 32-bit microcontroller). You cannot use other boards/controllers – unless you build one from discrete components (good luck with that!).
- 1.4. In addition to the Teensy and the tracked chassis components, you will be provided with:
 - 1.4.1. A box of parts including various sensors, actuators, and structural elements such as aluminium plate, rods, acrylic etc.
 - 1.4.2. A budget of \$50 which you can use to order additional components. (see 5. Procurement).
 - 1.4.3. 500g of PLA for 3D printing. Additional material will come out of your \$50 (1.4.2) at 5c/g
- 1.5. All robots need to **meet general safety standards**, including:
 - 1.5.1. Lasers must be **below 5mW power**, unless approved by Julian Phillips.
 - 1.5.2. Spinning devices must travel at **under 200rpm**, unless adequate guarding is in place.
 - 1.5.3. It is compulsory to **use the supplied power module, in between the battery and any electronics**.
 - 1.5.4. **No naked flames** allowed; this includes any form of flame thrower.
 - 1.5.5. **No chemically explosive or EMP devices**. :(
 - 1.5.6. Voltage within your device should not **exceed 100v DC**.
- 1.6. Rubber bands and other similar elastomeric components with a high aspect ratio (e.g. strings or strips) cannot be used on any part of your robot
 - 1.6.1. Elastomeric components are defined as polymers with high viscoelasticity allowing them to stretch significantly (>20% strain) under stress and return to their original shape when released.
 - 1.6.2. High aspect ratio (flat length/width) > 5.0.

2. Environment

- 2.1. The competition will take place in a **2.4 x 4.9m** meter arena with **400mm high walls**. The walls and floor of the arena will be coloured black.
- 2.2. Within the arena will be two areas designated as home base.
 - 2.2.1. These areas will be coloured blue and green.
 - 2.2.2. The base area is defined as the 600 x 600mm coloured space.
 - 2.2.3. There will be a low 'rim' attached to the floor within each base (**~10mm high**).
 - 2.2.4. The bases will be located in corners of the arena.
- 2.3. The arena will contain a number of obstacles that your robot will need to negotiate. Obstacles include walls, ramps and speed-bumps.
 - 2.3.1. All **gaps between walls** (including arena walls) **will be greater than 0.4m**. Robot size restrictions
 - 2.3.2. **Vertical obstacles** (e.g. walls and pipes) will be **coloured red**.
 - 2.3.3. **Horizontal obstacles** (speed bumps, ramps) will be the same colour as the arena floor.
 - 2.3.4. Speed-bumps will be rectangular in profile and **up to 25mm high**. Robot needs to be able to move over this
 - 2.3.5. Ramps **may be up to 100mm high**, with a **maximum gradient of 30%**.

3. Weights and mobile elements (*snitches and bludgers*)

- 3.1. Within the arena, there will be more than **5 target weights**.
- 3.2. All *target weights* will outwardly look identical.
 - 3.2.1. **Steel** cylinder of diameter 50mm and height 70mm with an annular groove to facilitate gripping.
 - 3.2.2. **Target weights will weigh 1.0kg, 0.75kg, or 0.5kg**.
- 3.3. Within the arena, there will be a number of *dummy weights*.
 - 3.3.1. *Dummy weights* will be the same size and shape as *target weights*, but constructed of **non-conducting plastic**.
 - 3.3.2. **Some dummy weights will have a steel insert in the top**. So maybe metal detector on side/bottom instead of top

Soft constraint: should be able to pick up in all orientations

- 3.4. **Weights may be placed against arena or obstacle walls.**
- 3.5. If a *weight* is knocked over during a round, it **will be left on its side** for the remainder of the round.
- 3.6. Within the arena there may be **up to 2 mobile elements** during the competition. *Mobile elements* consist of Sphero Bolt robots (<https://www.sphero.com/sphero-bolt>)
 - 3.6.1. Spherical plastic shell
 - 3.6.2. Diameter of approximately 73mm
 - 3.6.3. Speed of up to 7 m/s.
 - 3.6.4. LED matrix may be lit up to distinguish between elements:
 - 3.6.4.1. **Snitches will be illuminated yellow (gold).**
 - 3.6.4.2. **Bludgers will be illuminated red.**
 - 3.6.5. Path will be random.

Need a way to distinguish between colours

4. Competition

- 4.1. The competition consists of a number of rounds in which two robots are present in the arena simultaneously.
 - 4.1.1. Robots will begin each round on their respective base, **facing a direction specified by the competition director for each round.** (Should be able to swivel easily)
 - 4.1.2. The duration of each **round is 2 minutes.**
 - 4.1.3. At the end of each round, the team will **turn their robot off but will not lift it up.** The robots will be picked up by 'officials' and taken to the judge's desk for scoring – this is to prevent cheeky tilting of the robot to retain or drop weights.
- 4.2. The robot with the highest score at the end of the round will be declared the winner.
 - 4.2.1. The score for a robot is calculated at the end of the round by: $\text{Score} = 1x (\text{weight of target weights on board robot}) + 2x (\text{weight of target weights in home base at the end of the round}) + 3\text{kg} \times (\text{number of snitches on board robot}) - 3\text{kg} \times (\text{number of bludgers on board robot}) - 0.5\text{kg} \times (\text{number of dummy weights on board robot}) - \text{any penalties}$
- 4.3. A *weight/mobile element* will be considered **on board a robot when it is off the ground and completely under the control of the robot** i.e. if the robot is picked up off the ground, the *weights/mobile element* come with the robot. Weights must be secure when picked up to count, ie can't just drag it
- 4.4. *Weights* on board a robot are **not considered safe and can therefore be pilfered by the opposing team**, provided that no deliberate damage is inflicted during the raid (as per 4.18). Weights must be protected from stealing
- 4.5. Target weights that have been *delivered* to the team base are considered *safe* and cannot be *stolen* by the opposing robot.
 - 4.5.1. **A weight is considered delivered if it is on the ground touching any part of the coloured home base area**, and not on board/under the control of a robot.
 - 4.5.2. A weight is considered *stolen* if it is brought *on board* the opposing robot (as per 4.3)
 - 4.5.3. If one or more target weights is stolen from a base, those weights will still contribute to the final score of the team that originally gathered them and a penalty equivalent to this score will be subtracted from the thieving team. If it is unclear which weight(s) was stolen, it will be assumed to be the heaviest weight(s) among the target weights the thieving team has on board and in their home base.
 - 4.5.4. If one or more target weights are pushed outside the base without being stolen and are not returned by the end of the round, they will not count towards the score.
- 4.6. **Dummy weights** on board a robot subtract from the score (as per 4.2.1).
 - 4.6.1. *Dummy weights* that have been delivered to a team base (and are therefore not on board a robot) do not contribute to the score.
- 4.7. A robot can have a **maximum of 3 target weights on board** at the end of a round.
 - 4.7.1. Only 3 *target weights* can contribute to the on board part of the final score for the round.
 - 4.7.2. *Dummy weights* do not count towards the on board total. A robot can have many dummy weights on board – but they will count towards the final score.
 - 4.7.3. If a robot ends the round with more than 3 *target weights* on board, a penalty will be applied.
 - 4.7.3.1. For every additional target weight (>3) on board, a penalty of -1 points will be applied to the score at the end of the round.

- 4.7.3.2. Penalties will be limited so that the minimum score is 0 points.
- 4.7.3.3. If a robot finishes a round with >3 weights on board, the lightest target weights will contribute to the score, and the heaviest target weights will be omitted.
- 4.8. If two robots finish with the same score, the team with the greatest **cumulative weight successfully delivered to their home base will be declared the winner.**
- 4.9. If two robots finish with the same score and weight on their base, **the team with the greatest number of target weights (on board and at base) will be declared the winner** (via more successful pick-ups).
- 4.10. If two robots finish and are still tied according to 4.9, the team with the **fewest number of dummy weights will be declared the winner.**
- 4.11. In the unlikely event that two robots finish the round tied according to 4.10, the **lightest robot** will be declared the winner. *Soft constraint? but very low priority*
- 4.12. In the almost impossible event that two robots finish the round tied according to 4.11, the robot that has obviously travelled the greatest distance during the round (at the discretion of the competition director on the day) will be declared the winner. If one robot has not obviously travelled further than another, then the robot that is furthest from its base (in a straight line) will be declared the (dubious) winner.
- 4.13. In the 'this is just getting ridiculous' event that the round is tied according to 4.12, we'll flip a coin...
- 4.14. There will be **no restarts of robots.** If a robot malfunctions or is damaged accidentally, it must continue to the end of the round without intervention.
- 4.14.1. If both teams malfunction at the start of a round, the round may restart at the discretion of the competition director on the day.
- 4.14.2. If both robots malfunction/get stuck during a round, the competition director may offer the opportunity to restart or reposition the robots to continue to the round. Both teams must agree for this to occur.
- 4.15. **Robots can be repaired between rounds. However, they must be ready in time for their next round.**
- 4.16. If a robot is late to its round, it will be disqualified. This decision will be made at the discretion of the competition director. There will be no discussion or appeal.
- 4.17. Both robots must remain inside the arena at all times. If a **robot ventures outside the arena, whichever robot was responsible will be disqualified** (i.e. if a robot takes itself outside the arena, it will be disqualified. If one robot is placed outside the arena by the opposition, the opposition will be disqualified).
- 4.18. Any **deliberate damage to the other robot during a knockout round will result in immediate disqualification.** This is not Robot Wars! :(((
- 4.18.1. However, cunning methods to inhibit your competition that do not result in damage will be permitted. The decision on whether any damage is accidental or deliberate will be made by the competition director on the competition day – there will be no discussion or appeal, so be careful!
- 4.19. In the event that one **robot is captured by another, any weights on board the captured robot will be considered property of the captor.** *insane strat*
- 4.19.1. A robot is considered **captured** if no part of it is touching the ground and its movements are completely under the control of the captor – i.e. the captor must be able to move the captive robot at will and without restriction.
- 4.20. When the competition is completed, the design will be stripped down to the original components by the team and all parts returned to Julian – unless your robot is so damned awesome that you are requested to return the robot in one piece for a victory parade.
- 4.20.1. Robot strip down must be completed by the end of study week, else a penalty of 10% will be imposed on the team score for the final competition.
- 4.21. The competition will follow a double-elimination format, so a robot must lose 2 rounds before being eliminated.

5. Procurement

- 5.1. When spending your budget, **items must be ordered through Julian** and be able to be purchased using a credit card. If you buy any items yourself, or bring them from home, you **will not be reimbursed**.
- 5.2. Any items **not ordered through Julian will be assessed for value** (either on production of a receipt, or Julian's experience) and that amount deducted from your budget. This is not an all-out robot-beauty pageant, there are cost and hardware constraints, just like in the real-world.
 - 5.2.1. **Second-hand items will be valued at their current 'new' price** (if you were to buy that item today)
- 5.3. Procurement requests should be submitted to Julian via email and procurement will take place fortnightly, unless agreed in advance.
- 5.4. All materials used from the supplied parts kit or additional procurement shall be identified in a bill of materials in the design document.
- 5.5. No budget credit will be given for unused items in the supplied parts kit.

Reporting:

We have produced guidelines for each of the three reports that are required for this project. These guideline documents will be available on Learn. Basically, the structure of the three reports is:

1. Conceptual design report – research, present and, evaluate several possible concepts and select one to take forward
2. Design progress report – Present progress to date on the design, including some technical aspects such as engineering drawings, testing/verification results, fault tree analysis.
3. Design evaluation report - critically evaluate the performance of your final design and compare it against competing designs in the context of this competition.

These reports have strict page limits to ensure that you present only relevant and important information. The requirements for these reports have changed slightly from previous years and updated documents will be made available soon. Several examples of each type from previous years are available on Learn – these will differ slightly from what you will need to produce but will provide some inspiration.